

# Nicholas Le

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## EDUCATION

University of California, San Diego  
*B.S. in Computer Engineering*

Aug. 2024 – June 2028  
*San Diego, CA*

## WORK EXPERIENCE

### Undergraduate Research Assistant

Apr. 2026 – Present

*System Energy Efficiency (SEE) Lab, UC San Diego (PI: Prof. Tajana Rosing)*

*San Diego, CA*

- Reproduced the published HyperSpec GPU spectral-clustering benchmark (PXD000561, ~7.46M clusters) to within 0.001% of the reference run, after porting its **CUDA kernels** from Numba to **CuPy RawKernels** to enable execution on **NVIDIA Blackwell (sm\_120)** GPUs.
- Extended **HyperOMS**, a GPU open-modification search engine built on **Hyperdimensional Computing (HDC)**, to metabolomics — running spectral library search against ~1.5M positive-mode GNPS spectra and resolving a **ZIP64 npz/cnpy** binary incompatibility that had corrupted GPU-side hypervector encoding.
- Adapted the ANN-SoLo proteomics search tool to metabolomics and fixed a **LanceDB** query-explosion bug from non-unique GNPS identifiers, replacing a batch-by-identifier loop with an ordered sequential scan and cutting index construction from a failed state to seconds for ~1.5M spectra.
- Built a comparative pipeline benchmarking **HyperOMS** against **ANN-SoLo** on identical metabolomics queries, quantifying identification overlap to evaluate HDC-based search sensitivity versus cosine-based methods.

### Co-Founder & CEO

Jan. 2026 – Present

*Kyte - College Delivery Service*

*San Diego, CA*

- Co-founded and lead a college delivery startup, owning business strategy, go-to-market, and product roadmap for a multi-sided marketplace (**customer, runner, admin** clients) built for UCSD's **40,000-student** campus.
- Engineered a concurrent backend with **Next.js**, **TypeScript**, **PostgreSQL (Neon)**, and **Redis/BullMQ**, building a fault-tolerant transactional engine with strict payment idempotency.
- Built **real-time state management** and **live dispatch routing** for meetpoint-based fulfillment, using **schema-driven API contracts** to target a **sub-25-minute** delivery SLA.

### Software Engineering Intern

June 2025 – December 2025

*Health1st.ai — WinFully on Technologies*

*Remote (Alpharetta, GA)*

- Built **FastAPI microservices** handling **HL7, FHIR, and X12** healthcare data formats with async processing, **PostgreSQL**, and **Redis** caching, deployed to **GCP Cloud Run** via CI/CD.
- Developed an **AI mapping service** using **LangChain** and **GPT-4** to transform healthcare data between standard formats (HL7, FHIR, X12, CCDA).

## PROJECTS

### Custom 9-bit RISC Processor | *SystemVerilog, Quartus II, ModelSim, FPGA, Assembly*

Spring 2026

- Designed a custom **9-bit fixed-length ISA** 8-bit datapath, 8 registers, R/S/I instruction formats optimized for three target programs under a tight instruction-width budget.
- Implemented the CPU in **SystemVerilog** (register file, ALU, control decoder, data memory, fetch unit) and wrote a matching **assembler** (assembly → 9-bit machine code), verified with **ModelSim** and synthesized in **Quartus II** for an **Intel Cyclone IV E** FPGA.
- Hand-wrote assembly for Hamming-distance, min/max-difference, and **signed 16×16 → 32-bit shift-and-add multiply** programs, building 16-/32-bit arithmetic from single-byte ALU ops via carry/borrow flags.

### NBA Player Props EV Engine | *Python, NumPy, SciPy, Pandas*

Dec. 2025

- Engineered a **Monte Carlo simulation engine** (20K+ iterations) utilizing a **Student-t copula** ( $\nu = 5$ ) to capture fat-tailed tail dependence, and applied Higham's algorithm for nearest-PSD correlation matrices.
- Modeled discrete prop probabilities via **Poisson and Negative Binomial distributions**, integrating a logit-space **Bayesian market blender** to anchor empirical projections with sharp sportsbook implied probabilities.

## SKILLS AND COURSEWORK

**Languages:** Python, C, C++, Java, JavaScript, TypeScript, Swift, Bash, HTML/CSS

**Hardware & Systems:** SystemVerilog, Verilog, CUDA, FPGA (Intel Cyclone IV), Quartus II, ModelSim/Questa, Computer Architecture, Digital Design

**Frameworks/Libraries:** Next.js, FastAPI, React, Node.js, Prisma, Redis, NumPy, Pandas, SciPy, FAISS, LanceDB, Docker, PostgreSQL, MongoDB, AWS S3

**Relevant Courses:** Data Structures & OOP (CSE 12), Discrete Math (CSE 20), Systems Prog (CSE 29), Comp Architecture (CSE 30), Computer Architecture Lab (CSE 141L), Circuit Analysis (ECE 35 & 140), Digital Systems (CSE 140), Probability & Stats (ECE 109)